Manual Chapter : Implementing Health and Performance Monitoring

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Implementing Health and Performance Monitoring

Overview: Health and performance monitoring

You can set up the BIG-IP system to monitor the health or performance of certain nodes or servers that are members of a load balancing pool. Monitors verify connections on pool members and nodes. A monitor can be either a health monitor or a performance monitor, designed to check the status of a pool, pool member, or node on an ongoing basis, at a set interval. If a pool member or node being checked does not respond within a specified timeout period, or the status of a pool member or node indicates that performance is degraded, the BIG-IP system can redirect the traffic to another pool member or node.

Some monitors are included as part of the BIG-IP system, while other monitors are user-created. Monitors that the BIG-IP system provides are called pre-configured monitors. User-created monitors are called custom monitors.

Before configuring and using monitors, it is helpful to understand some basic concepts regarding monitor types, monitor settings, and monitor implementation.

Monitor types
Every monitor, whether pre-configured or custom, is a certain type of monitor. Each type of monitor checks the status of a particular protocol, service, or application. For example, one type of monitor is HTTP. An HTTP type of monitor allows you to monitor the availability of the HTTP service on a pool, pool member, or node. A WMI type of monitor allows you to monitor the performance of a pool, pool member, or node that is running the Windows Management Instrumentation (WMI) software. An ICMP type of monitor simply determines whether the status of a node is up or down.

Monitor settings
Every monitor consists of settings with values. The settings and their values differ depending on the type of monitor. In some cases, the BIG-IP system assigns default values. For example, the following shows the settings and default values of an ICMP-type monitor.

```
Name my_icmp
Type ICMP
Interval 5
Timeout 16
Transparent No
Alias Address * All Addresses
```

If you want to monitor the performance of a RealNetworks® RealServer server or a Windows-based server equipped with Windows Management Instrumentation (WMI), you must first download a special plug-in file onto the BIG-IP system.

Task summary for implementing health and performance monitoring

To implement a health or performance monitor, you perform these tasks.

Creating a custom monitor

Before creating a custom monitor, you must decide on a monitor type.

You can create a custom monitor when the values defined in a pre-configured monitor do not meet your needs, or no pre-configured monitor exists for the type of monitor you are creating.

When defining values for custom monitors, make sure you avoid using any values that are on the list of reserved keywords.

1. On the Main tab, click **Local Traffic › Monitors .**
   The Monitors List screen opens.
2. Click **Create.**
   The New Monitor screen opens.
3. In the **Name** field, type a name for the monitor.
4. From the **Type** list, select the type of monitor.
   The screen refreshes, and displays the configuration options for the monitor type.
5. From the **Import Monitor** list, select an existing monitor.

   The new monitor inherits initial configuration values from the existing monitor.
6. From the **Configuration** list, select **Advanced.**

   This selection makes it possible for you to modify additional default settings.
7. Configure all settings shown.
8. Click **Finished.**

Creating a load balancing pool

Ensure that at least one virtual server exists in the configuration before you start to create a load balancing pool.

Create a pool of systems with Access Policy Manager to which the system can load balance global traffic.

1. On the Main tab, click **DNS › GSLB › Pools.**
   The Pool List screen opens.
2. Click **Create.**
   The New Pool screen opens.
3. In the General Properties area, in the **Name** field, type a name for the pool.

   Names must begin with a letter, and can contain only letters, numbers, and the underscore (_) character.

   The pool name is limited to 63 characters.
4. From the **Type** list, depending on the type of the system (IPv4 or IPv6), select either an **A** or **AAAA** pool type.
5. In the Configuration area, for the **Health Monitors** setting, in the **Available** list, select a monitor type, and move the monitor to the **Selected** list.

   Hold the Shift or Ctrl key to select more than one monitor at a time.
6. In the Members area, for the **Load Balancing Method** settings, select a method that uses virtual server score:
   - VS Score - If you select this method, load balancing decisions are based on the virtual server score only.
   - Quality of Service - If you select this method, you must configure weights for up to nine measures of service, including **VS Score.** Virtual server score then factors into the load balancing decision at the weight you specify.
7. For the **Member List** setting, add virtual servers as members of this load balancing pool.

   The system evaluates the virtual servers (pool members) in the order in which they are listed. A virtual server can belong to more than one pool.

   a. Select a virtual server from the **Virtual Server** list.

   b. Click **Add.**
8. Click **Finished.**

Creating a virtual server

A virtual server represents a destination IP address for application traffic.

1. On the Main tab, click **Local Traffic › Virtual Servers .**
   The Virtual Server List screen opens.
2. Click **Create.**
   The New Virtual Server screen opens.
3. In the **Name** field, type a unique name for the virtual server.
4. In the **Destination Address/Mask** field:
   - If you want to specify a single IP address, confirm that the **Host** button is selected, and type the IP address in CIDR format.
   - If you want to specify multiple IP addresses, select the **Address List** button, and confirm that the address list that you previously created appears in the box.

   The supported format is address/prefix, where the prefix length is in bits. For example, an IPv4 address/prefix is `10.0.0.1` or `10.0.0.0/24`, and an IPv6 address/prefix is `ffe1::0020/64` or `2001:ed8:77b5:2:10:10:100:42/64`. When you use an IPv4 address without specifying a prefix, the BIG-IP system automatically uses a `/32` prefix.

   The IP address or addresses for this field must be on the same subnet as the external self-IP address.
5. From the **Service Port** list, select ***All Ports.**
6. In the Resources area of the screen, from the **Default Pool** list, select the relevant pool name.

You now have a destination IP address on the BIG-IP system for application traffic.

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